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ANOMALY DETECTION IN OPERATIONAL TECHNOLOGY ENVIRONMENT

Abstract

Approaches for automated and efficient detection of anomalies in operational technology (OT) environments are described. According to one example, real-time operation data corresponding to an entity operating within an OT environment of an organization may be obtained. The real-time operation data may be indicative of operations performed by the entity within the OT environment. For each of the operations, operational information associated with the operation may be identified. The real-time operation data and the operational information may be processed, utilizing the anomaly detection model, to detect any anomaly in the operations. One or more preventive actions may be initiated within the OT environment upon detecting an anomaly in at least one of the operations.

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Background/Summary

BACKGROUND

[0001] Generally, across all industries, numerous operations are performed on a daily basis in a controlled operational environment. For example, in industrial sectors, such as oil, gas, mining, chemicals, energy, manufacturing, and defense, the industries include an operational technology (OT) environment for monitoring and controlling physical industrial processes and for taking business decisions such as for scheduling of production, for use of material, for shipping, etc. The OT environment may include processing equipment and field devices, such as sensors and actuators, which perform physical processes of the industries. Further, the OT environment may include devices for managing production workflows and instruments for sending commands to the processing equipment and field devices. Furthermore, the OT environment may include an industrial control system (ICS) such as a distributed control system (DCS) or a supervisory control and data acquisition (SCADA) system for supervising, monitoring, and controlling the physical processes.

[0002] With large scale digitalization, most of the industrial processes are also being automated to enhance operational efficiency and enable data-driven decision-making and remote management. For such automation, the OT environment has become increasingly interconnected with wired and wireless networks, including the Internet, to collect, analyze, and leverage data on industry's premises and in the cloud. Thus, the OT environments have become increasingly exposed to cyber threats that may compromise the safety and reliability of the industrial operations.

Description

BRIEF DESCRIPTION OF FIGURES

[0003] Systems and/or methods are now described, in accordance with examples of the present subject matter and with reference to the accompanying figures, in which:

[0004] FIG. 1 illustrates a system for detecting an anomaly in an operational technology (OT) environment, according to an example;

[0005] FIG. 2A and FIG. 2B illustrate a computing environment implementing the system for detecting an anomaly in an OT environment, according to another example;

[0006] FIG. 3 illustrates a data flow diagram for detecting an anomaly in an OT environment, according to an example;

[0007] FIG. 4 illustrates a method for detecting an anomaly in an OT environment, according to an example;

[0008] FIG. 5A to FIG. 5C illustrate a method for training of a machine learning model for detecting an anomaly in an OT environment, according to an example;

[0009] FIG. 6A to FIG. 6C illustrate a method for detecting an anomaly in an OT environment, according to another example; and

[0010] FIG. 7 illustrates a computing environment implementing a non-transitory computer-readable medium for detecting an anomaly in an OT environment, according to an example.

DETAILED DESCRIPTION

[0011] OT environments are vital for effective operation of industrial processes. The connectivity of the OT environments to wired or wireless networks exposes the OT environments to cyber threats that may compromise the safety and reliability of industrial operations. Cyber-attacks

directed at systems or devices within the OT environments may result in an unauthorized access of critical industrial data, data breaches, interruption of crucial processes, and monetary losses. Inefficient cybersecurity for the OT environments of an organization may thus result in undesired operational disruptions, system failures, and downtime, thereby leading to severe consequences, including production delays, decreased efficiency of the OT environment, reputational damage for the organization, and financial losses for the organization.

[0012] Inadequate OT cybersecurity may further cause safety risks to employees of the organization, the public, and the environment. For example, cyber-attacks targeting the OT environments in industries such as manufacturing, energy, transportation, etc., may potentially lead to dangerous accidents, equipment malfunctions, or environmental disasters, jeopardizing human lives and causing significant damage to the environment and the organization's infrastructure. Such accidents or disasters may even lead to non-compliance of standard regulations due to which the organization may suffer regulatory penalties, legal repercussions, and reputational harm. Inefficient cybersecurity for the OT environments may put the organization at a competitive disadvantage due to lack of trust in customers, partners, and other stakeholders.

[0013] Once the security of the OT environment is breached, addressing vulnerabilities in the OT cybersecurity may be expensive. For example, the organization may need to cover expenses for the loss in productivity due to the downtime, court costs, fines, customer compensation, and damage control costs, and even compensation to cyber attackers involved in the breach. Thus, organizations implement security measures for prevention of cyber-attacks on the OT environments.

[0014] Traditional security measures make use of pre-defined rules for monitoring and filtering network traffic which is being received at the OT environment and which is going out of the OT environment. Such pre-defined rules are typically created once and are then used for a long time. Such pre-defined rules are generally created based on known type of cyber-attacks.

[0015] However, such traditional security measures prove to be insufficient against detection of any anomaly in the OT environment. This is because the number and sophistication of cyber-attacks is continuously increasing day by day. Moreover, it is challenging for the organizations to stay updated on the most recent threats and vulnerabilities because of the scattered information, lack of centralized/right expertise, and limited sharing of latest threats among organizations. As a result, the OT environments become more susceptible to cyber-attacks if the traditional security measures are continued.

[0016] Moreover, there may be numerous users within the organization. A specific user of the organization may have authorization to monitor and control a specific industrial operation while another user may have authorization to monitor and control another industrial operation. Existing techniques for detecting cyber-attacks fail to effectively identify if a particular action is an anomaly or not. For example, the existing techniques fail to identify if the particular action is initiated by an authorized user, a malicious user, or an unauthorized user. Therefore, there is a need for security measures which can efficiently prevent cyber-attacks on the OT environments.

[0017] The present subject matter describes approaches for automated and efficient detection of anomalies in operational technology (OT) environments. The approaches include analyzing real-time operation data corresponding to an entity operating within an OT environment of an organization. In an example, the entity may be an asset such as a device, a system, or a machine associated with the organization. In another example, the entity may be a user operating one or more assets associated with the organization. The real-time operation data may be indicative of one or more operations performed by the entity within the OT environment. Operations performed by each entity operating within the OT environment may be individually monitored in real-time to detect any anomaly in the operations. In an example, behavior-based monitoring may be implemented to detect performance of any operation that is not typically performed by the entity while operating within the OT environment. In addition or alternatively, role-based monitoring may be implemented to detect performance of any operation that the entity is not authorized to perform

according to the role and responsibilities assigned to that entity. Upon detecting an anomaly in at least one of the operations, one or more preventive actions may be initiated within the OT environment for preventing occurrence of any undesirable event within the OT environment. The claimed invention utilizes a generative artificial intelligence (AI) model that may be referred to as an anomaly detection model for implementing the behavior-based monitoring and the role-based monitoring. The anomaly detection model may be utilized in conjunction with behavior analytics for tracking behavior of an entity over time, thereby, enabling detection of anomalies in real-time, so that appropriate action may be timely taken to prevent occurrence of any unwanted circumstances. The described approaches thus provide a simple and robust analytical methodology for early, quick, efficient, and automated detection of anomalies in the OT environment. Further, the anomaly detection model may facilitate in providing a dynamic and adaptive cybersecurity system.

[0018] In an example implementation of the present subject matter, for enabling the role-based monitoring, historical role-specific activity data may be analyzed to obtain an initial version of the anomaly detection model. The historical role-specific activity data may indicate ideal operations performed within one or more OT environments of one or more organizations, by a plurality of authorized entities that is authorized to perform operations corresponding to roles assigned in at least one of the one or more organizations. Thus, the anomaly detection model may be trained to understand what operations are normally performed by entities assigned with different roles and responsibilities across organizations.

[0019] In an example implementation of the present subject matter, for enabling the behaviour-based monitoring, historical operation data corresponding to one or more entities associated with the organization may be obtained. The historical operation data may be indicative of one or more historical operations performed by each of the one or more entities within the OT environment of the organization. For each of the one or more historical operations, a particular historical time at which the historical operation was performed may be identified. The historical operation data may then be analyzed in correlation with the particular historical time to obtain a final trained version of the anomaly detection model. Thus, the anomaly detection model may be trained to understand an ideal operational behavior of each of the one or more entities over a period of time.

[0020] In an example, for implementing the behavior-based monitoring for an entity using the anomaly detection model, timing information may be identified for each of one or more operations performed by the entity. The timing information may indicate a particular time at which the operation was performed. Further, a historical behaviour-based pattern of the entity may be obtained. The historical behaviour-based pattern may be indicative of historical operations performed by the entity at the particular time. The historical operations may be compared with the one or more operations to detect the anomaly. The anomaly may be detected upon detecting a deviation of at least one operation of the one or more operations from the historical operations.

[0021] In an example, for implementing the role-based monitoring for an entity using the anomaly detection model, role-based access control (RBAC) information may be identified for each of one or more operations performed by the entity. The RBAC information may indicate authorization details and responsibilities assigned to the entity for operating within the OT environment. Further, a historical role-based pattern may be obtained for the entity. The historical role-based pattern may be indicative of ideal operations performed by an ideal entity having authorization to operate within an ideal OT environment according to the authorization details and responsibilities. The ideal operations may be compared with the one or more operations to detect the anomaly. The anomaly may be detected upon detecting a deviation of at least one operation of the one or more operations from the ideal operations.

[0022] In an example, the one or more preventive actions that may be initiated upon detecting an anomaly in at least one of the one or more operations may include generation of a suspension signal for transmission to one or more devices associated with the organization. The suspension

signal may prevent execution of the one or more operations for which the anomaly is detected. Further, the one or more preventive actions may include generation of an alert notification for transmission to a supervisor on a supervisor device. The alert notification may be indicative of the anomaly, enabling the supervisor to proactively engage in adversary pursuit and threat hunting. [0023] Since the anomaly detection model is obtained through training on the historical role-specific activity data and the historical operation data, the described approaches may enable easy and quick detection of anomalies within the OT environment by identifying role-based or behavior-based deviations in operations performed by an entity within the OT environment. As a result, cyber-attacks on the OT environments may be prevented without a need for organizations to stay updated on the most recent types of cyber-threats and vulnerabilities.

[0024] The described approaches utilize advanced behavioral analytics and machine learning platforms to efficiently process high volume of historical data from organizations for detecting the anomalies in real-time. Any person or machine that interacts with a company, system, platform, plant assets or product of an organization can be a subject of the behavioral analytics. The described approaches make the organizations capable of collating and analyzing operations from various IT/OT sources in real-time to identify potential issues before such operations can impact the OT environment. That is, a high volume of raw event data including operations from interactions across multiple channels is ingested, looking at everything from time to time, enabling quick detection of the anomalies. This proactive detection reduces the likelihood of successful cyberattacks, reducing the amount of time for which an adversary is in the OT environment, and minimizes business disruption. Upon detecting an anomaly, a corrective measure may be taken immediately and any suspicious activity may be immediately arrested at the moment of detection. Thus, the described approaches provide a comprehensive protection against cyber-attacks and enable a robust cybersecurity for the OT environment which enhances the reputation of the organization and the trust in customers, partners, and other stakeholders. Thus, the organization may be protected from safety hazards and covering expenses which would have otherwise been required to be covered in case of any undesired event. Further, the described approaches help the organizations to avoid reputational damage, regulatory penalties, legal repercussions, or jeopardizing the customer's lives.

[0025] The present subject matter is further described with reference to FIG. 1 to FIG. 7. It should be noted that the description and figures merely illustrate principles of the present subject matter. Various arrangements may be devised that, although not explicitly described or shown herein, encompass the principles of the present subject matter. Moreover, all statements herein reciting principles, aspects, and examples of the present subject matter, as well as specific examples thereof, are intended to encompass equivalents thereof.

[0026] FIG. 1 illustrates a system **100** for detecting an anomaly in an operational technology (OT) environment, according to an example. In one example, the system **100** may be a distributed computing system having one or more physical computing systems geographically distributed at same or different locations. In another example, one or more components of the system **100** may be hosted virtually, for example, on a cloud-based platform, while other components may be geographically distributed at same or different locations. In yet another example, the system **100** may be a stand-alone physical system geographically located at a particular location. In an example, the system **100** may be utilized by organizations that aim to secure their OT environments from cyber-attacks.

[0027] In one example, the system **100** may include engine(s) **102** and data **104**. The system **100** may also include additional components, such as display, input/output interfaces, operating systems, applications, and other software or hardware components (not shown in the figures).

[0028] The engine(s) **102** may be implemented as a combination of hardware and programming, for example, programmable instructions to implement a variety of functionalities of the engine(s) **102**. In examples described herein, such combinations of hardware and programming may be

implemented in several different ways. For example, the programming for the engine(s) **102** may be executable instructions. Such instructions may be stored on a non-transitory machine-readable storage medium which may be coupled either directly with the system **100** or indirectly (for example, through networked means). In an example, the engine(s) **102** may include a processing resource, for example, either a single processor or a combination of multiple processors, to execute such instructions. In the present examples, the non-transitory machine-readable storage medium may store instructions that, when executed by the processing resource, implement the engine(s) **102**. In other examples, the engine(s) **102** may be implemented as electronic circuitry.

[0029] In one example, the engine(s) **102** may include a data acquisition engine **106**, an anomaly detection engine **108**, an OT security engine **110**, and other engine(s) **112**. The other engine(s) **112** may further implement functionalities that supplement functions performed by the system **100** or any of the engine(s) **102**.

[0030] The data **104** includes data that is either received, stored, or generated as a result of functions implemented by any of the engine(s) **102** or the system **100**. It may be further noted that information stored and available in the data **104** may be utilized by the engine(s) **102** for performing various functions of the system **100**. The data **104** may include real-time operation data **114** and other data **116**. The real-time operation data **114** may be indicative of one or more operations performed, in real-time, within the OT environment of an organization hosting the system **100**. The other data **116** may include data that is either received, stored, or generated as a result of functions implemented by any of the engine(s) **102**.

[0031] In operation, the data acquisition engine **106** may obtain real-time operation data corresponding to an entity operating within an OT environment of an organization. The real-time operation data may be indicative of one or more operations performed by the entity within the OT environment. In an example, the entity may be an asset such as a device, a system, or a machine associated with the organization. In another example, the entity may be a user operating one or more assets associated with the organization. In an example, the real-time operation data may be obtained from the asset operating within the OT environment. In another example, the real-time operation data may be obtained from a centralized server managing operations performed within the OT environment of the organization. In one example, the real-time operation data may be stored as the real-time operation data **114**.

[0032] Once the real-time operation data is obtained, for each of the one or more operations, the anomaly detection engine **108** may identify operational information associated with the operation. In an example, the operational information may comprise at least one of timing information and role-based access control (RBAC) information. The timing information may indicate a particular time at which the operation was performed. Further, the RBAC information may indicate authorization details and responsibilities assigned to the entity for operating within the OT environment. For example, the organization may have users which are assigned with respective roles such as “operator”, “engineer”, and “supervisor”. According to the role assigned by the organization, the entity may have a pre-defined set of responsibilities and access permissions which may hereinafter be referred to as the authorization details and responsibilities. In an example, the anomaly detection engine **108** may implement an anomaly detection model to identify the operational information. The anomaly detection model may be a generative artificial intelligence (AI) model trained on historical data to detect anomalies within the OT environment.

[0033] The anomaly detection engine **108** may process the real-time operation data and the operational information to detect any anomaly in the one or more operations. In an example, the anomaly detection engine **108** may utilize the anomaly detection model to process the real-time operation data and the operational information. In an example, an anomaly may be detected whenever any operation in the one or more operations is detected that is not typically performed by the entity at the particular time while operating within the OT environment. In addition, or alternatively, an anomaly may be detected whenever any operation in the one or more operations is

detected that does not correspond to the authorization details and responsibilities assigned to the entity.

[0034] Subsequently, the OT security engine **110** may initiate one or more preventive actions within the OT environment upon detecting an anomaly in at least one of the one or more operations. In an example, the one or more preventive actions may include alerting a supervisor about the anomaly so that the supervisor may proactively engage in adversary pursuit and threat hunting. In an example, the one or more preventive actions may include controlling one or more devices associated with the organization to prevent execution of the one or more operations for which the anomaly is detected. Thus, the described approaches provide a simple and robust analytical methodology for early, quick, efficient, and automated detection of anomalies in the OT environment. Further, the anomaly detection model may facilitate in providing a dynamic and adaptive cybersecurity system.

[0035] FIG. 2A and FIG. 2B illustrate a computing environment **200** implementing the system **100** for detecting an anomaly in an OT environment **202**, according to another example. In one example, the computing environment **200** may include the system **100**, the OT environment **202**, and a supervisor device **204**.

[0036] In an example, the OT environment **202** may be associated with a particular organization. The OT environment **202** may include OT assets **208** and OT users **210**. The OT assets **208** may be assets **208-1**, . . . , **208-N** belonging to the organization, where N may be a natural number. The assets **208-1**, . . . , **208-N** may be individually referred to as asset **208** and collectively referred to as the OT assets **208**. The asset **208** may be a processing equipment, a field device, an electronic device, a system, or any machine operating within the OT environment of the organization. Physical processes of the organization, production workflows of the organization, and control parameters for processing equipment or field devices operating within the OT environment **202** may be controlled through the OT assets **208**. In an example, the asset **208** may be a processing equipment or a field device, such as a sensor or an actuator, which performs physical industrial processes of the organization. In another example, the asset **208** may be a device for managing production workflows. In yet another example, the asset **208** may be an instrument for sending commands to the processing equipment or the field device. In yet another example, the asset **208** may be an industrial control system (ICS) such as a distributed control system (DCS) or a supervisory control and data acquisition (SCADA) system for supervising, monitoring, and controlling the physical processes. As exemplarily illustrated in FIG. 2B, examples of the asset **208** may include, but are not limited to, a sensor **208-1**, a computer **208-2**, a server **208-3**, a printing machine **208-4**, a camera **208-5**, and a laptop **208-6**, operating within the OT environment. The sensor **208-1** may be any type of sensor, such as a temperature sensor and a pressure sensor. The server **208-3** may store and manage data associated with the organization and the assets **208**. Although only hardware components have been illustrated as the OT assets **208** in FIG. 2B, it should be understood that the OT assets **208** may also include software assets utilized by the organization for implementing various industrial processes. In an example, the asset **208** may perform one or more operations based on direct interaction with at least one of the OT users **210**. In another example, the asset **208** may perform one or more operations without direct interaction with the OT users **210**.

[0037] The OT users **210** may be users **210-1**, . . . , **210-M** associated with the organization, where M may be a natural number. The users **210-1**, . . . , **210-M** may be individually referred to as user **210** and collectively referred to as the OT users **210**. The user **210** may interact with at least one of the OT assets **208** to cause the asset **208** to perform one or more operations. FIG. 2B exemplarily illustrates user A **210-1**, user B **210-2**, user C **210-3**, and user D **210-4**. Although four OT users **210** have been illustrated as an example, it should be understood that the OT environment may include any number of the OT users **210**. The user **210** may be assigned a corresponding role within the organization. The corresponding role may be associated with pre-defined responsibilities and pre-

defined permissions to interact and access the OT assets **208**. For example, user A **210-1** may be assigned permission to control some features in the laptop **208-6**, while user B **210-2** may be assigned permission to read data managed by the server **208-3**. Further, user C **210-3** may be responsible to control the operation of the printing machine **208-4** based on data generated by the sensor **208-1**, while user D may be responsible to control the operation of the camera **208-5** using the computer **208-2**. Similarly different users **210** may be assigned respective responsibilities and respective permissions for accessing and controlling one or more of the OT assets **208**.

[0038] In an example, the supervisor device **204** may be a device over which the system **100** may provide notification to a user, such as a supervisor of an organization, about anomalies detected within an OT environment of the organization. The supervisor device **204** may be accessed by the supervisor associated with the organization. In an example, the supervisor may access the supervisor device **204** to receive alerts regarding the anomalies. As exemplarily illustrated in FIG. 2B, examples of the supervisor device **204** may include, but are not limited to, a laptop **204-1** and a mobile phone **204-2**. Examples of the supervisor device **204** may also include, but are not limited to, a desktop, a tablet computer, a personal digital assistant (PDA) and any electronic device capable of transmitting or receiving data. Although one supervisor device **204** has been illustrated in FIG. 2A and two supervisor devices **204-1** and **204-2** have been illustrated in FIG. 2B for the sake of brevity, it should be understood to a person skilled in the art that any number of supervisor devices **204** may be connected with the system **100** to receive alerts about the anomalies.

[0039] The system **100**, the OT environment **202**, and the supervisor device **204** may be communicably coupled with each other over a communication network **206** and may exchange data and signals over the communication network **206**. The communication network **206** may be a wireless network, a wired network, or a combination thereof. The communication network **206** may also be an individual network or a collection of many such individual networks, interconnected with each other and functioning as a single large network, e.g., the Internet or an intranet. Examples of such individual networks include local area network (LAN), wide area network (WAN), the internet, Global System for Mobile Communication (GSM) network, Universal Mobile Telecommunications System (UMTS) network, Personal Communications Service (PCS) network, Time Division Multiple Access (TDMA) network, Code Division Multiple Access (CDMA) network, Next Generation Network (NGN), Public Switched Telephone Network (PSTN), and Integrated Services Digital Network (ISDN).

[0040] Depending on the technology, the communication network **206** may include various network entities, such as transceivers, gateways, and routers. In an example, the communication network **206** may include any communication network that uses any of the commonly used protocols, for example, Hypertext Transfer Protocol (HTTP), and Transmission Control Protocol/Internet Protocol (TCP/IP).

[0041] In one example, the system **100** may include processor(s) **212**, interface(s) **214**, memory **216**, a communication module **218**, the engine(s) **102**, and the data **104**. The system **100** may also include other components, such as display, input/output interfaces, operating systems, applications, and other software or hardware components (not shown in the figures).

[0042] The processor(s) **212** may be implemented as microprocessors, microcomputers, microcontrollers, digital signal processors, central processing units, state machines, logic circuitries, and/or other devices that manipulate signals based on operational instructions. The interface(s) **214** may allow the connection or coupling of the system **100** with one or more other devices, such as the supervisor device **204** and the OT assets **208** within the OT environment **202**, through a wired (e.g., Local Area Network, i.e., LAN) connection or through a wireless connection (e.g., Bluetooth®, Wi-Fi). The interface(s) **214** may also enable intercommunication between different logical as well as hardware components of the system **100**.

[0043] The memory **216** may be a computer-readable medium, examples of which include volatile memory (e.g., RAM), and/or non-volatile memory (e.g., Erasable Programmable read-only

memory, i.e., EPROM, flash memory, etc.). The memory **216** may be an external memory or an internal memory, such as a flash drive, a compact disk drive, an external hard disk drive, or the like. The memory **216** may further include the data **104** and/or other data which may either be received, utilized, or generated during the operation of the system **100**.

[0044] The communication module **218** may be a wireless communication module. Examples of the communication module **218** may include, but are not limited to, Global System for Mobile communication (GSM) modules, Code-division multiple access (CDMA) modules, Bluetooth modules, network interface cards (NIC), Wi-Fi modules, dial-up modules, Integrated Services Digital Network (ISDN) modules, Digital Subscriber Line (DSL) modules, and cable modules. In one example, the communication module **218** may also include one or more antennas to enable wireless transmission and reception of data and signals. The communication module **218** may allow the system **100** to transmit data and signals to one or more other devices, such as the supervisor device **204** and the OT assets **208** within the OT environment **202**; and receive data and signals from the one or more other devices.

[0045] The engine(s) **102** may include the data acquisition engine **106**, the anomaly detection engine **108**, the OT security engine **110**, and the other engine(s) **112**, as explained with reference to FIG. 1. In an example, the engine(s) **102** may further include a model training engine **220**.

[0046] The data **104** may include the real-time operation data **114** and the other data **116**, as explained with reference to FIG. 1. In an example, the data **104** may further include entity data **222**, historical role-specific activity data **224**, and historical operation data **226**. In an example, the entity data **222** may include role-based access control (RBAC) information for each entity, such as the asset **208** and the user **210**, operating within the OT environment **202**. The RBAC information may define role and responsibility of the entity within the organization. That is, the RBAC information may indicate authorization details and responsibilities assigned to the entity for operating within the OT environment **202**. In an example, the entity data **222** may be obtained from the OT assets **208** or the OT users **210** operating within the OT environment **202**. In another example, the entity data **222** may be pre-stored in the memory **216** of the system **100**. The historical role-specific activity data **224** may be historical data indicating ideal operations performed within one or more OT environments of one or more organizations, by a plurality of authorized entities authorized to perform operations corresponding to roles assigned in at least one of the one or more organizations. In an example, the historical role-specific activity data **224** may be ideal operations generally performed according to each of a plurality of pre-defined roles across different organizations. In an example, the historical role-specific activity data **224** may be industry-specific ideal operations performed according to each of a plurality of pre-defined roles across different organizations belonging to a particular industry. For example, the historical role-specific activity data **224** may indicate ideal operations performed by engineers in a manufacturing industry. The historical operation data **226** may correspond to one or more entities associated with the organization. The historical operation data **226** may be indicative of one or more historical operations performed by each of the one or more entities within the OT environment **202** of the organization. In an example, the historical operation data **226** may include historical timing information for each of the one or more historical operations. The historical timing information may indicate a particular historical time at which the historical operation was performed. In an example, the historical operation data **226** may be obtained from the OT assets **208** or the OT users **210** operating within the OT environment **202**.

[0047] In operation, for enabling detection of anomalies in operations performed within the OT environment **202**, the model training engine **220** of the system **100** may be configured to train an anomaly detection model. The anomaly detection model may be a generation AI model that can identify patterns in data provided for training to use such patterns for detection of the anomalies in real-time.

[0048] In an example, for training the anomaly detection model, the model training engine **220** may

obtain historical role-specific activity data. The historical role-specific activity data may indicate ideal operations performed within one or more OT environments of one or more organizations, by a plurality of authorized entities authorized to perform operations corresponding to roles assigned in at least one of the one or more organizations. In an example, the historical role-specific activity data may indicate ideal operations generally performed according to each of a plurality of pre-defined roles across different organizations. In an example, the historical role-specific activity data may indicate industry-specific ideal operations performed according to each of the plurality of pre-defined roles across different organizations belonging to a particular industry. Examples of the plurality of pre-defined roles may include, but are not limited to, an operator, an engineer, and a supervisor. For example, the historical role-specific activity data may indicate ideal operations performed by engineers in a manufacturing industry. In an example, the historical role-specific activity data may be pre-stored in the historical role-specific activity data **224**.

[0049] Once the historical role-specific activity data is obtained, the model training engine **220** may analyze the historical role-specific activity data to obtain an initial version of the anomaly detection model. In an example, the initial version of the anomaly detection model may be utilized to implement role-based monitoring of operations performed within the OT environment **202**. The role-based monitoring may involve detecting any particular operation that is performed by a particular entity which is not authorized to perform that particular operation according to the role and responsibilities assigned to the particular entity.

[0050] In an example, for analyzing the historical role-specific activity data, the model training engine **220** may identify corresponding role-based access control (RBAC) details for each of the plurality of authorized entities. The corresponding RBAC details may indicate authorization details and responsibilities assigned to the authorized entity for operating within the one or more OT environments as per the roles assigned to the authorized entity in at least one of the one or more organizations.

[0051] Subsequently, the model training engine **220** may identify a historical role-based pattern for each of the plurality of authorized entities. The historical role-based pattern may be indicative of a correlation between the ideal operations and the corresponding RBAC details. Thus, the historical role-based pattern provides a typical pattern of operations performed by users that are assigned a particular role within the one or more organizations. For example, the historical role-based pattern may indicate what operations are typically performed by engineers within one or more organizations.

[0052] The model training engine **220** may then train the anomaly detection model based on the historical role-based pattern to obtain the initial version of the anomaly detection model.

[0053] In addition, or alternatively, for training the anomaly detection model, the model training engine **220** may obtain historical operation data corresponding to one or more entities associated with an organization for which the anomaly detection model is to be utilized for detection of anomalies. The historical operation data may be indicative of one or more historical operations performed by each of the one or more entities within the OT environment **202** of the organization. In an example, the historical operation data may include historical timing information for each of the one or more historical operations. The historical timing information may be indicative of a particular historical time at which the historical operation was performed. In an example, the historical operation data may be obtained from the OT assets **208** or the OT users **210** operating within the OT environment **202**. In an example, the historical operation data may be pre-stored in the historical operation data **226**.

[0054] Once the historical operation data is obtained, for each of the one or more historical operations, the model training engine **220** may identify the particular historical time at which the historical operation was performed. Further, the model training engine **220** may analyze the historical operation data in correlation with the particular historical time to obtain a final trained version of the anomaly detection model. The final trained version of the anomaly detection model

may be utilized to implement behavior-based monitoring of operations performed within the OT environment **202** of the organization for which the final trained version of the anomaly detection model is obtained. The behavior-based monitoring may involve detecting any particular operation that is performed by an entity that does not typically perform that particular operation at the time at which the particular operation is performed.

[0055] In an example, for analyzing the historical operation data, the model training engine **220** may identify a historical behaviour-based pattern for the entity. The historical behaviour-based pattern may be indicative of a correlation between the one or more historical operations and the particular historical time. Thus, the historical behaviour-based pattern provides a typical pattern of how assets **208** within the OT environment **202** operate at different times. For example, the historical behaviour-based pattern may indicate what operations are performed by the asset **208** at a particular time. For instance, actuators within the OT environment may operate for one hour and then rest without operating for five minutes. Similar patterns may be detected for the assets **208** operating within the OT environment **202** in correlation with the time of operation.

[0056] The model training engine **220** may then optimize the initial version of the anomaly detection model based on the historical behaviour-based pattern to obtain the final trained version of the anomaly detection model.

[0057] For utilizing the anomaly detection model to detect anomalies in real-time, the data acquisition engine **106** may obtain real-time operation data corresponding to an entity operating within the OT environment **202** of the organization. The real-time operation data may be indicative of one or more operations performed by the entity within the OT environment **202**. In an example, the entity may be the asset **208** associated with the organization. In another example, the entity may be the user **210** operating the asset **208** associated with the organization. In an example, the real-time operation data may be obtained from the OT assets **208** operating within the OT environment **202**. In another example, the real-time operation data may be obtained from a centralized server, say the server **208-3**, managing operations performed within the OT environment **202** of the organization. The real-time operation data may be obtained for real-time monitoring of the one or more operations to enable detection of the anomalies. In one example, the real-time operation data may be stored as the real-time operation data **114**.

[0058] Once the real-time operation data is obtained, for each of the one or more operations, the anomaly detection engine **108** may identify operational information associated with the operation. In an example, the operational information may comprise at least one of timing information and role-based access control (RBAC) information. The timing information may indicate a particular time at which the operation was performed. In an example, the real-time operation data include a time-stamp tag corresponding to each of the one or more operations and the timing information may accordingly be identified based on the time-stamp tag. Further, the RBAC information may indicate authorization details and responsibilities assigned to the entity for operating within the OT environment **202**. For example, the organization may have users **210** which are assigned with respective roles such as “operator”, “engineer”, and “supervisor”. According to the role assigned by the organization, the entity may have a pre-defined set of responsibilities and access permissions which may hereinafter be referred to as the authorization details and responsibilities. In an example, the anomaly detection engine **108** may implement the anomaly detection model to identify the operational information. In an example, the RBAC information may be stored in the entity data **222**.

[0059] The anomaly detection engine **108** may process the real-time operation data and the operational information to detect any anomaly in the one or more operations. In an example, the anomaly detection engine **108** may utilize the anomaly detection model to process the real-time operation data and the operational information. In an example, an anomaly may be detected whenever any operation in the one or more operations is detected that is not typically performed by the entity at the particular time while operating within the OT environment **202**. In addition, or

alternatively, an anomaly may be detected whenever any operation in the one or more operations is detected that does not correspond to the authorization details and responsibilities assigned to the entity.

[0060] In an example, for processing the real-time operation data and the operational information, the anomaly detection engine **108** may obtain the historical behaviour-based pattern of the entity. The historical behaviour-based pattern may be indicative of historical operations performed by the entity at the particular time.

[0061] Subsequently, the anomaly detection engine **108** may compare the historical operations with the one or more operations to detect the anomaly. In an example, the anomaly may be detected when a deviation of at least one operation of the one or more operations from the historical operations is detected. Thus, by comparing the historical operations with the one or more operations, the anomaly detection model may enable implementation of the behavior-based monitoring for detecting anomalies related to typical behavior of the entity.

[0062] In an example, for processing the real-time operation data and the operational information, the anomaly detection engine **108** may obtain a historical role-based pattern for the entity. The historical role-based pattern may be indicative of ideal operations performed by an ideal entity having authorization to operate within an ideal OT environment according to the authorization details and responsibilities. The ideal operations may be authentic operations, the ideal entity may be an authentic entity, and the ideal OT environment may be an authentic OT environment. Thus, the historical role-based pattern may be indicative of authentic operations that should be performed at a respective role assigned to the authentic entity.

[0063] Subsequently, the anomaly detection engine **108** may compare the ideal operations with the one or more operations to detect the anomaly. In an example, the anomaly may be detected when a deviation of at least one operation of the one or more operations from the ideal operations is detected. Thus, by comparing the ideal operations with the one or more operations, the anomaly detection model may enable implementation of the role-based monitoring for detecting anomalies related to the role assigned to the entity within the organization.

[0064] In an example, the OT security engine **110** of the system **100** may initiate one or more preventive actions within the OT environment **202** upon detecting an anomaly in at least one of the one or more operations.

[0065] In an example, the one or more preventive actions may include alerting a supervisor about the anomaly so that the supervisor may proactively engage in adversary pursuit and threat hunting. For alerting the supervisor, the OT security engine **110** may generate an alert notification for transmission to the supervisor on the supervisor device **204**. The alert notification may be indicative of the anomaly.

[0066] In an example, the one or more preventive actions may include controlling the OT assets **208** to prevent execution of the one or more operations for which the anomaly is detected. For controlling the OT assets **208**, the OT security engine **110** may generate a suspension signal for transmission to one or more devices associated with the organization. The one or more devices may be any of the OT assets **208**. The suspension signal may be to prevent execution of the one or more operations for which the anomaly is detected. For example, if a malicious user is trying to copy and paste some confidential data to an external memory using the laptop **208-6** which is typically operated by the user B **210-2**, the anomaly detection engine may detect, based on the behavior-based monitoring, that user B **210-2** typically does not try to copy and paste the confidential data. In addition or alternatively, if user B **210-2** is not authorized to copy and paste the confidential data according to the role and responsibilities assigned to the user B **210-2**, the anomaly detection engine may detect, based on the role-based monitoring, that user B **210-2** is not authorized to copy and paste the confidential data. Accordingly, the OT security engine **110** may generate the alert notification and the communication module **218** may transmit the alert notification to the supervisor device **204** informing about the attempt to copy and paste the confidential data. Further,

the OT security engine **110** may generate the suspension signal and the communication module **218** of the system **100** may transmit the suspension signal to the laptop **208-6**. The laptop **208-6** may disallow pasting of the confidential data into the external memory. Thus, the described approaches provide a simple and robust analytical methodology for early, quick, efficient, and automated detection of anomalies in the OT environment. Further, the anomaly detection model may facilitate in providing a dynamic and adaptive cybersecurity system.

[0067] FIG. **3** illustrates a data flow diagram **300** for detecting an anomaly in an OT environment, say the OT environment **202**, according to an example. The order in which the data flow diagram **300** is described is not intended to be construed as a limitation, and some of the described components of the data flow diagram **300** may be combined in a different order to implement a data flow according to the data flow diagram **300**, or an alternative data flow.

[0068] The data flow in the data flow diagram **300** may be implemented in a suitable hardware, computer-readable instructions, or combination thereof. The steps of such data flow diagram **300** may be performed by either a system under the instruction of machine executable instructions stored on a non-transitory computer-readable medium or by dedicated hardware circuits, microcontrollers, or logic circuits. For example, the data flow in the data flow diagram **300** may be performed by components of the system **100**. In an implementation, the data flow of the data flow diagram **300** may be performed under an “as a service” delivery model, where the system **100** operated by a provider, receives programmable code. In an implementation, the data flow of the data flow diagram **300** and the system **100** may be implemented within a demilitarized zone (DMZ) which separates the OT environment from a business and logistics zone of an organization. The demilitarized zone is typically considered as level 3.5 in a Purdue Model usually employed by industries as a reference model for data flows. Herein, some examples are also intended to cover non-transitory computer-readable medium, for example, digital data storage media, which are computer-readable and encode computer-executable instructions, where said instructions perform some or all the steps of the data flow of the data flow diagram **300**.

[0069] In one example implementation, the data flow diagram **300** of FIG. **3** illustrates historical role-specific activity data **302-1** and **302-2**. The historical role-specific activity data **302-1** and **302-2** may be the historical role-specific activity data **224** explained with reference to FIG. **2A**. In an example, the historical role-specific activity data **224** may be divided into the historical role-specific activity data **302-1** and the historical role-specific activity data **302-2**. The historical role-specific activity data **302-1** may be utilized for training of an anomaly detection model. The historical role-specific activity data **302-1** may be utilized for testing of the anomaly detection model.

[0070] The data flow diagram **300** illustrates a block **304** for data normalization and pre-processing. At block **304**, the historical role-specific activity data **302-1** may be normalized and pre-processed.

[0071] The data flow diagram **300** illustrates a block **306** for model training. The historical role-specific activity data **302-1** after normalization and pre-processing, and the historical role-specific activity data **302-2** may be fed to the block **306** for model training. In an example, the historical role-specific activity data **302-2** may also be normalized and pre-processed before being fed to the block **306** for model training.

[0072] The block **306** for model training includes a rule engine **308** and an error state identifier **310**. The rule engine **308** may recognize the historical role-based patterns for one or more entities using the normalized and pre-processed historical role-specific activity data. A historical role-based pattern for an entity may indicate typical patterns in operational behavior of the entity using the normalized and pre-processed historical ICS data. In an example, the typical patterns may be recognized based on how and what activities the entity typically perform according to role and responsibilities assigned to the entity for operating within the OT environment. In another example, the typical patterns may be recognized based on how and what data the entity typically accesses or

modifies according to role and responsibilities assigned to the entity for operating within the OT environment. Based on the typical patterns, the rule engine **308** may create rules for categorizing a particular action taken by a particular entity as one of a legitimate action and a malicious action for detection of the anomaly.

[0073] The error state identifier **310** may test the rules created by the rule engine **308** based on the historical role-specific activity data **302-2** to identify errors in the rules. The error state identifier **310** may improvise the rules according to the identified errors to generate an initial version **312** of the anomaly detection model. In an example, the anomaly detection model may be a generative AI model. In an example, the anomaly detection model may be a convolutional neural networks (CNN) model. In another example, the anomaly detection model may be a recurrent neural network (RNN) model.

[0074] The data flow diagram **300** illustrates historical operation data **314**. The historical operation data **314** maybe same as the historical operation data **226**, explained with reference to FIG. 2A. In an example, the historical operation data **314** may define data based on which the OT environment is typically operated. For example, the historical operation data **314** may include control parameters used for the operation of the OT assets **208** of FIG. 2A, the order of the operation of the OT assets **208**, etc.

[0075] The data flow diagram **300** further illustrates a block **316** for fine tuning. At block **316**, the initial version **312** of the anomaly detection model may be fine-tuned using the historical operation data **314** to generate a final trained version **318** of the anomaly detection model. The anomaly detection model may be utilized, in real-time, for detecting anomalies in the OT environment whenever any action is performed by any entity associated with the organization.

[0076] The data flow diagram **300** further illustrates a block **320** for anomaly detection in real-time which is fed with the final trained version **318** of the anomaly detection model. The block **320** further illustrates real-time operation data **322**. The real-time operation data **322** may be same as the real-time operation data **114**, explained with reference to FIG. 1 and FIG. 2A.

[0077] The block **320** further illustrates a block **324** for processing. At block **324**, the real-time operation data **322** may be processed utilizing the anomaly detection model to generate result **326**, illustrated in FIG. 3, regarding anomalies within the OT environment. At the block **324** for processing, the real-time operation data **322** may also be normalized and pre-processed before generating the result **326** utilizing the anomaly detection model. In an example, the real-time operation data **322** may be processed in the same manner as explained with reference to FIG. 2A to generate the result **326**. The result **326** may indicate whether any anomaly is detected in one or more operations indicated by the real-time operation data **322**. Based on the result **326**, appropriate preventive actions may be initiated, as explained with reference to FIG. 2A.

[0078] FIG. 4, FIG. 5A, FIG. 5B, FIG. 5C, FIG. 6A, FIG. 6B, and FIG. 6C illustrate example methods **400**, **500**, **504**, **510**, **600**, **606**, and **606**, respectively, for detecting an anomaly in an OT environment and training of a machine learning model for detecting an anomaly in an OT environment. The order in which the methods are described is not intended to be construed as a limitation, and any number of the described method blocks may be combined in any order to implement the methods, or an alternative method. Further, the methods **400**, **500**, **504**, **510**, **600**, and **606** may be implemented by processing resource or computing device(s) through any suitable hardware, non-transitory machine-readable instructions, or combination thereof.

[0079] It may also be understood that methods **400**, **500**, **504**, **510**, **600**, and **606** may be performed by programmed computing devices, such as the system **100**, as depicted in FIG. 1, FIG. 2A, and FIG. 2B. Furthermore, the methods **400**, **500**, **504**, **510**, **600**, and **606** may be executed based on instructions stored in a non-transitory computer-readable medium, as will be readily understood. The non-transitory computer-readable medium may include, for example, digital memories, magnetic storage media, such as one or more magnetic disks and magnetic tapes, hard drives, or optically readable digital data storage media. While the methods **400**, **500**, **504**, **510**, **600**, and **606**

are described below with reference to the system **100** as described above; other suitable systems for the execution of these methods may also be utilized. Additionally, implementation of the methods **400**, **500**, **504**, **510**, **600**, and **606** is not limited to such examples.

[0080] FIG. **4** illustrates the method **400** for detecting an anomaly in an OT environment of an organization, according to an example.

[0081] At block **402**, real-time operation data corresponding to an asset, say the asset **208**, operating within the OT environment, say the OT environment **202**, may be obtained. The real-time operation data may be indicative of one or more operations performed by the asset. In an example, the asset may be a device, a system, or a machine associated with the organization. In an example, the real-time operation data may be obtained from the asset. In another example, the real-time operation data may be obtained from a centralized server, say the server **208-3**, managing operations performed within the OT environment of the organization.

[0082] At block **404**, role-based access control (RBAC) information may be obtained. The RBAC information may be indicative of authorization details and responsibilities assigned to a user, say the user **210**, operating the asset. For example, the organization may have users which are assigned with respective roles such as “operator”, “engineer”, and “supervisor”. According to the role assigned by the organization, the user may have a pre-defined set of responsibilities and access permissions which may be referred to as the authorization details and responsibilities.

[0083] At block **406**, the real-time operation data and the RBAC information may be processed to detect any anomaly in the one or more operations. In an example, the real-time operation data and the RBAC information may be processed utilizing an anomaly detection model. The anomaly detection model may be a generative artificial intelligence (AI) model trained on historical data to detect anomalies within the OT environment. In an example, an anomaly may be detected whenever any operation in the one or more operations is detected that does not correspond to the authorization details and responsibilities assigned to the user.

[0084] At block **408**, it is determined whether an anomaly is detected in at least one of the one or more operations. If any anomaly is not detected in the one or more operations, the method may move back to block **402** and the real-time operation data may be continuously obtained and processed.

[0085] Upon detecting an anomaly in at least one of the one or more operations, at block **410**, one or more preventive actions may be initiated within the OT environment. In an example, the one or more preventive actions may include alerting a supervisor about the anomaly so that the supervisor may proactively engage in adversary pursuit and threat hunting. In an example, the one or more preventive actions may include controlling one or more devices associated with the organization to prevent execution of the one or more operations for which the anomaly is detected. Thus, the described approaches provide a simple and robust analytical methodology for early, quick, efficient, and automated detection of anomalies in the OT environment. Further, the anomaly detection model may facilitate in providing a dynamic and adaptive cybersecurity system.

[0086] FIG. **5A** illustrates the method **500** for training of a machine learning model for detecting an anomaly in an OT environment of an organization, according to an example.

[0087] At block **502**, historical role-specific activity data may be obtained. The historical role-specific activity data may indicate ideal operations performed within one or more OT environments of one or more organizations, by a plurality of authorized entities authorized to perform operations corresponding to roles assigned in at least one of the one or more organizations. In an example, the historical role-specific activity data may indicate ideal operations generally performed according to each of a plurality of pre-defined roles across different organizations. In an example, the historical role-specific activity data may indicate industry-specific ideal operations performed according to each of the plurality of pre-defined roles across different organizations belonging to a particular industry. Examples of the plurality of pre-defined roles may include, but are not limited to, an operator, an engineer, and a supervisor. For example, the historical role-specific activity data may

indicate ideal operations performed by engineers in a manufacturing industry.

[0088] At block **504**, the historical role-specific activity data may be analyzed to obtain an initial version of an anomaly detection model. The anomaly detection model may be the machine learning model. In an example, the initial version of the anomaly detection model may be utilized to implement role-based monitoring of operations performed within the OT environment. The role-based monitoring may involve detecting any particular operation that is performed by a particular entity which is not authorized to perform that particular operation according to the role and responsibilities assigned to the particular entity.

[0089] In addition or alternatively, at block **506**, historical operation data corresponding to one or more entities associated with an organization may be obtained. The organization may be a particular organization for which the anomaly detection model is to be utilized for detection of anomalies. The historical operation data may be indicative of one or more historical operations performed by each of the one or more entities within the OT environment of the organization. In an example, the historical operation data may include historical timing information for each of the one or more historical operations. The historical timing information may be indicative of a particular historical time at which the historical operation was performed. In an example, the historical operation data may be obtained from assets, say the OT assets **208**, or users, say the OT users **210**, operating within the OT environment.

[0090] At block **508**, for each of the one or more historical operations, identify the particular historical time may be identified at which the historical operation was performed.

[0091] At block **510**, the historical operation data may be analyzed in correlation with the particular historical time to obtain a final trained version of the anomaly detection model. The final trained version of the anomaly detection model may be utilized to implement behavior-based monitoring of operations performed within the OT environment of the organization for which the final trained version of the anomaly detection model is obtained. The behavior-based monitoring may involve detecting any particular operation that is performed by an entity that does not typically perform that particular operation at the time at which the particular operation is performed.

[0092] FIG. **5B** illustrates the method **504** for analyzing the historical role-specific activity data at block **504** of FIG. **5A**, according to an example. In an example, for each of the plurality of authorized entities, corresponding RBAC details may indicate authorization details and responsibilities assigned to the authorized entity for operating within the one or more OT environments as per the roles assigned to the authorized entity in at least one of the one or more organizations.

[0093] At block **512**, a historical role-based pattern may be identified for each of the plurality of authorized entities. The historical role-based pattern may be indicative of a correlation between the ideal operations and the corresponding RBAC details. Thus, the historical role-based pattern provides a typical pattern of operations performed by users that are assigned a particular role within the one or more organizations. For example, the historical role-based pattern may indicate what operations are typically performed by engineers within one or more organizations.

[0094] At block **514**, the anomaly detection model may be trained based on the historical role-based pattern to obtain the initial version of the anomaly detection model.

[0095] FIG. **5C** illustrates the method **510** for analyzing the historical operation data at block **510** of FIG. **5A**, according to an example.

[0096] At block **516**, a historical behaviour-based pattern may be identified for the asset. The historical behaviour-based pattern may be indicative of a correlation between the one or more historical operations and the particular historical time. Thus, the historical behaviour-based pattern provides a typical pattern of how assets within the OT environment operate at different times. For example, the historical behaviour-based pattern may indicate what operations are performed by the asset at a particular time. For instance, actuators within the OT environment may operate for one hour and then rest without operating for five minutes. Similar patterns may be detected for the

assets operating within the OT environment in correlation with the time of operation.

[0097] At block **518**, the initial version of the anomaly detection model may be optimized based on the historical behaviour-based pattern to obtain the final trained version of the anomaly detection model.

[0098] FIG. **6** illustrates the method **600** for detecting an anomaly in an OT environment of an organization, according to another example.

[0099] At block **602**, real-time operation data corresponding to an entity operating within the OT environment may be obtained. The real-time operation data may be indicative of one or more operations performed by the entity within the OT environment. In an example, the entity may be an asset, say the asset **208**, associated with the organization. In another example, the entity may be a user, say the user **210**, operating the asset associated with the organization. In an example, the real-time operation data may be obtained from OT asset, say the OT assets **208**, operating within the OT environment. In another example, the real-time operation data may be obtained from a centralized server, say the server **208-3**, managing operations performed within the OT environment of the organization. The real-time operation data may be obtained for real-time monitoring of the one or more operations to enable detection of the anomalies.

[0100] At block **604**, for each of the one or more operations, operational information associated with the operation may be identified. In an example, the operational information may comprise at least one of timing information and role-based access control (RBAC) information. The timing information may indicate a particular time at which the operation was performed. In an example, the real-time operation data include a time-stamp tag corresponding to each of the one or more operations and the timing information may accordingly be identified based on the time-stamp tag. Further, the RBAC information may indicate authorization details and responsibilities assigned to the entity for operating within the OT environment. For example, the organization may have users which are assigned with respective roles such as “operator”, “engineer”, and “supervisor”. According to the role assigned by the organization, the entity may have a pre-defined set of responsibilities and access permissions which may be referred to as the authorization details and responsibilities. In an example, the anomaly detection model, obtained through training in FIGS. **5A**, **5B**, and **5C**, may be implemented to identify the operational information.

[0101] At block **606**, the real-time operation data and the operational information may be processed to detect any anomaly in the one or more operations. In an example, the anomaly detection model may be utilized to process the real-time operation data and the operational information. In an example, an anomaly may be detected whenever any operation in the one or more operations is detected that is not typically performed by the entity at the particular time while operating within the OT environment. In addition or alternatively, an anomaly may be detected whenever any operation in the one or more operations is detected that does not correspond to the authorization details and responsibilities assigned to the entity.

[0102] At block **608**, it is determined whether an anomaly is detected in at least one of the one or more operations. If any anomaly is not detected in the one or more operations, the method may move back to block **602** and the real-time operation data may be continuously obtained and processed.

[0103] Upon detecting an anomaly in at least one of the one or more operations, one or more preventive actions may be initiated within the OT environment. In an example, the one or more preventive actions may include controlling one or more devices associated with the organization to prevent execution of the one or more operations for which the anomaly is detected. For instance, at block **610**, a suspension signal may be generated for transmission to one or more devices associated with the organization. The one or more device may be any of the OT assets. The suspension signal may be to prevent execution of the one or more operations for which the anomaly is detected. For example, if a malicious user is trying to copy and paste some confidential data to an external memory using a particular laptop which is typically operated by a particular authentic user,

based on the behavior-based monitoring, it may be detected that the particular authentic user typically does not try to copy and paste the confidential data. In addition or alternatively, if the particular authentic user is not authorized to copy and paste the confidential data according to the role and responsibilities assigned to the particular authentic user, based on the role-based monitoring, it may be detected that the particular authentic user is not authorized to copy and paste the confidential data. Further, the suspension signal may be generated and transmitted to the particular. The particular laptop may disallow pasting of the confidential data into the external memory.

[0104] In an example, the one or more preventive actions may include alerting a supervisor about the anomaly so that the supervisor may proactively engage in adversary pursuit and threat hunting. For instance, at block **612**, an alert notification may be generated for transmission to the supervisor on a supervisor device. The alert notification may be indicative of the anomaly. For instance, the alert notification may be generated and transmitted to the supervisor device informing about the attempt to copy and paste the confidential data. Thus, the described approaches provide a simple and robust analytical methodology for early, quick, efficient, and automated detection of anomalies in the OT environment.

[0105] FIG. **6B** illustrates the method **606** for processing the real-time operation data and the operational information, say the timing information, at block **606** of FIG. **6A**, according to an example.

[0106] At block **614**, obtain the historical behaviour-based pattern of the entity. The historical behaviour-based pattern may be indicative of historical operations performed by the entity at the particular time.

[0107] At block **616**, the historical operations may be compared with the one or more operations to detect the anomaly. In an example, the anomaly may be detected when a deviation of at least one operation of the one or more operations from the historical operations is detected. Thus, by comparing the historical operations with the one or more operations, the anomaly detection model may enable implementation of the behavior-based monitoring for detecting anomalies related to typical behavior of the entity.

[0108] FIG. **6C** illustrates the method **606** for processing the real-time operation data and the operational information, say the RBAC information, at block **606** of FIG. **6A**, according to an example.

[0109] At block **618**, a historical role-based pattern may be obtained for the entity. The historical role-based pattern may be indicative of ideal operations performed by an ideal entity having authorization to operate within an ideal OT environment according to the authorization details and responsibilities. The ideal operations may be authentic operations, the ideal entity may be an authentic entity, and the ideal OT environment may be an authentic OT environment. Thus, the historical role-based pattern may be indicative of authentic operations that should be performed at a respective role assigned to the authentic entity.

[0110] At block **620**, the ideal operations may be compared with the one or more operations to detect the anomaly. In an example, the anomaly may be detected when a deviation of at least one operation of the one or more operations from the ideal operations is detected. Thus, by comparing the ideal operations with the one or more operations, the anomaly detection model may enable implementation of the role-based monitoring for detecting anomalies related to the role assigned to the entity within the organization.

[0111] FIG. **7** illustrates a computing environment **700** implementing a non-transitory computer-readable medium for detecting an anomaly in an OT environment, according to an example. In an example, the computing environment **700** includes processor(s) **702** communicatively coupled to a non-transitory computer-readable medium **704** through a communication link **706**. In one example, the communication link **706** may be similar to the communication network **206**, as described in conjunction with the preceding figures. In an example implementation, the computing environment

700 may be for example, the computing environment **200**. In an example, the processor(s) **702** may have one or more processing resources for fetching and executing computer-readable instructions from the non-transitory computer-readable medium **704**. The processor(s) **702** and the non-transitory computer-readable medium **704** may be implemented, for example, in the system **100** (as has been described in conjunction with the preceding figures).

[0112] The non-transitory computer-readable medium **704** may be, for example, an internal memory device or an external memory device. In an example implementation, the communication link **706** may be a network communication link. The processor(s) **702** and the non-transitory computer-readable medium **704** may also be communicatively coupled to the OT environment **202** over a network **708**. The network **708** may be similar to the communication network **206** described in conjunction with FIG. 2.

[0113] In an example implementation, the non-transitory computer-readable medium **704** may include a set of computer-readable instructions **710** which may be accessed by the processor(s) **702** through the communication link **706**. Referring to FIG. 7, in an example, the non-transitory computer-readable medium **704** may include instructions **710** that may cause the processor(s) **702** to obtain real-time operation data corresponding to a user functioning within the OT environment of an organization. The real-time operation data may be indicative of one or more operations performed by the user, say the user **210**, within the OT environment, say the OT environment **202**, of the organization. In an example, the real-time operation data may be obtained from one or more assets, say the assets **208**, operating within the OT environment. In another example, the real-time operation data may be obtained from a centralized server, say the server **208-3**, managing operations performed within the OT environment of the organization.

[0114] In an example, for each of the one or more operations, the instructions **710** may further cause the processor(s) **702** to identify timing information associated with the operation. The timing information may indicate a particular time at which the operation was performed.

[0115] In one example, the instructions **710** may cause the processor(s) **702** to process the real-time operation data and the timing information to detect any deviation of the one or more operations from a historical operating pattern of the user in terms of the particular time. In an example, the real-time operation data and the timing information may be processed utilizing an anomaly detection model. The anomaly detection model may be a generative artificial intelligence (AI) model trained on historical data to detect anomalies within the OT environment. In an example, the historical operating pattern may be generated using the anomaly detection model. The historical operating pattern may be indicative of historical operations performed by the user at the particular time.

[0116] In one example, the instructions **710** may cause the processor(s) **702** to initiate one or more preventive actions within the OT environment upon detecting a deviation of at least one of the one or more operations from the historical operating pattern.

[0117] In an example, the one or more preventive actions may include alerting a supervisor about the deviation so that the supervisor may proactively engage in adversary pursuit and threat hunting. For alerting the supervisor, the instructions **710** may cause the processor(s) **702** to generate an alert notification for transmission to the supervisor on a supervisor device. The alert notification may be indicative of the at least one operation.

[0118] In an example, the one or more preventive actions may include controlling OT assets, say the OT assets **208**, to prevent execution of the at least one operation for which the deviation is detected. For controlling the OT assets, the instructions **710** may cause the processor(s) **702** to generate a suspension signal for transmission to one or more devices associated with the organization. The one or more device may be any of the OT assets. The suspension signal may be to prevent execution of the at least one operation.

[0119] In an example, the anomaly detection model may be trained for utilizing for detecting the deviation. In an example, for training the anomaly detection model, the instructions **710** may cause

the processor(s) **702** to obtain, for the organization, historical operation data corresponding to one or more entities associated with the organization. The historical operation data may be indicative of one or more historical operations performed by each of the one or more entities within the OT environment of the organization. In an example, the historical operation data may include historical timing information for each of the one or more historical operations. The historical timing information may indicate a particular historical time at which the historical operation was performed.

[0120] Subsequently, for each of the one or more historical operations, the instructions **710** may cause the processor(s) **702** to identify the particular historical time at which the historical operation was performed.

[0121] The instructions **710** may then cause the processor(s) **702** to analyze the historical operation data in correlation with the particular historical time to obtain the anomaly detection model. The anomaly detection model may be utilized to implement behavior-based monitoring of operations performed within the OT environment of the organization for which the anomaly detection model is obtained. The behavior-based monitoring may involve detecting any particular operation that is performed by a user that does not typically perform that particular operation at the time at which the particular operation is performed.

[0122] In an example, to analyze the historical operation data, the instructions **710** may then cause the processor(s) **702** to identify a historical behaviour-based pattern for the user. The historical behaviour-based pattern may be indicative of a correlation between the one or more historical operations and the particular historical time. Thus, the historical behaviour-based pattern provides a typical pattern of how users operate within the OT environment at different times. For example, the historical behaviour-based pattern may indicate what operations are performed by the user at a particular time.

[0123] Subsequently, the instructions **710** may then cause the processor(s) **702** to analyze the historical behaviour-based pattern to obtain the anomaly detection model.

[0124] In an example, for processing the real-time operation data and the timing information, the instructions **710** may cause the processor(s) **702** to obtain the historical behaviour-based pattern of the user. The historical behaviour-based pattern may be indicative of historical operations performed by the user at the particular time. Further, the instructions **710** may cause the processor(s) **702** to compare the historical operations with the one or more operations to detect the deviation.

[0125] Thus, the described approaches provide a simple and robust analytical methodology for early, quick, efficient, and automated detection of anomalies in the OT environment. Further, the anomaly detection model may facilitate in providing a dynamic and adaptive cybersecurity system.

[0126] Although examples for the present disclosure have been described in language specific to structural features and/or methods, it is to be understood that the appended claims are not necessarily limited to the specific features or methods described. Rather, the specific features and methods are disclosed and explained as examples of the present disclosure.

Claims

1. A system comprising: a data acquisition engine to: obtain real-time operation data corresponding to an entity operating within an operational technology (OT) environment of an organization, the real-time operation data being indicative of one or more operations performed by the entity within the OT environment of the organization; an anomaly detection engine implementing an anomaly detection model to: for each of the one or more operations, identify operational information associated with the operation, the operational information comprising at least one of timing information and role-based access control (RBAC) information, wherein the timing information indicates a particular time at which the operation was performed, and wherein the RBAC

information indicates authorization details and responsibilities assigned to the entity for operating within the OT environment; and process, utilizing the anomaly detection model, the real-time operation data and the operational information to detect any anomaly in the one or more operations; and an OT security engine to initiate one or more preventive actions within the OT environment upon detecting an anomaly in at least one of the one or more operations.

2. The system of claim 1, wherein the entity is one of an asset associated with the organization and a user operating the asset.

3. The system of claim 1, wherein the system comprises a model training engine to: obtain historical role-specific activity data indicating ideal operations performed within one or more OT environments of one or more organizations, by a plurality of authorized entities authorized to perform operations corresponding to roles assigned in at least one of the one or more organizations; analyze the historical role-specific activity data to obtain an initial version of the anomaly detection model; obtain, for the organization, historical operation data corresponding to one or more entities associated with the organization, wherein the historical operation data is indicative of one or more historical operations performed by each of the one or more entities within the OT environment of the organization; for each of the one or more historical operations, identify a particular historical time at which the historical operation was performed; and analyze the historical operation data in correlation with the particular historical time to obtain a final trained version of the anomaly detection model.

4. The system of claim 3, wherein for each of the plurality of authorized entities, corresponding RBAC details indicate authorization details and responsibilities assigned to the authorized entity for operating within the one or more OT environments as per the roles assigned to the authorized entity in at least one of the one or more organizations, and wherein to analyze the historical role-specific activity data, the model training engine is to: identify a historical role-based pattern for each of the plurality of authorized entities, wherein the historical role-based pattern is indicative of a correlation between the ideal operations and the corresponding RBAC details; and train the anomaly detection model based on the historical role-based pattern to obtain the initial version of the anomaly detection model.

5. The system of claim 3, wherein to analyze the historical operation data, the model training engine is to: identify a historical behaviour-based pattern for the entity, wherein the historical behaviour-based pattern is indicative of a correlation between the one or more historical operations and the particular historical time; and optimize the initial version of the anomaly detection model based on the historical behaviour-based pattern to obtain the final trained version of the anomaly detection model.

6. The system of claim 1, wherein the one or more preventive actions include at least one of: generating a suspension signal for transmission to one or more devices associated with the organization, wherein the suspension signal is to prevent execution of the one or more operations for which the anomaly is detected; and generating an alert notification for transmission to a supervisor on a supervisor device, wherein the alert notification is indicative of the anomaly.

7. The system of claim 1, wherein to process the real-time operation data and the operational information, the anomaly detection engine is to: obtain a historical behaviour-based pattern of the entity, wherein the historical behaviour-based pattern is indicative of historical operations performed by the entity at the particular time; and compare the historical operations with the one or more operations to detect the anomaly, wherein detecting the anomaly comprises detecting a deviation of at least one operation of the one or more operations from the historical operations.

8. The system of claim 1, wherein to process the real-time operation data and the operational information, the anomaly detection engine is to: obtain a historical role-based pattern for the entity, wherein the historical role-based pattern is indicative of ideal operations performed by an ideal entity having authorization to operate within an ideal OT environment according to the authorization details and responsibilities; and compare the ideal operations with the one or more

operations to detect the anomaly, wherein detecting the anomaly comprises detecting a deviation of at least one operation of the one or more operations from the ideal operations.

9. A method comprising: obtaining real-time operation data corresponding to an asset operating within an operational technology (OT) environment of an organization, the real-time operation data being indicative of one or more operations performed by the asset; obtaining role-based access control (RBAC) information indicative of authorization details and responsibilities assigned to a user operating the asset; processing, utilizing an anomaly detection model, the real-time operation data and the RBAC information to detect any anomaly in the one or more operations; and initiating one or more preventive actions within the OT environment upon detecting an anomaly in at least one of the one or more operations.

10. The method of claim 9, wherein the method comprises: for each of the one or more operations, obtaining timing information indicative of a particular time at which the operation was performed; processing, utilizing the anomaly detection model, the real-time operation data and the timing information to detect any other anomaly in the one or more operations; and initiating one or more additional preventive actions within the OT environment upon detecting another anomaly in at least one of the one or more operations.

11. The method of claim 9, wherein the method comprises: obtaining historical role-specific activity data indicating ideal operations performed within one or more OT environments of one or more organizations, by a plurality of authorized entities authorized to perform operations corresponding to roles assigned in at least one of the one or more organizations; analyzing the historical role-specific activity data to obtain an initial version of the anomaly detection model; obtaining, for the organization, historical operation data corresponding to one or more assets associated with the organization, wherein the historical operation data is indicative of one or more historical operations performed by each of the one or more assets within the OT environment of the organization; for each of the one or more historical operations, identifying a particular historical time at which the historical operation was performed; and analyzing the historical operation data in correlation with the particular historical time to obtain a final trained version of the anomaly detection model.

12. The method of claim 11, wherein for each of the plurality of authorized entities, corresponding RBAC details indicate authorization details and responsibilities assigned to the authorized entity for operating within the one or more OT environments as per the roles assigned to the authorized entity in at least one of the one or more organizations, and wherein analyzing the historical role-specific activity data comprises: identifying a historical role-based pattern for each of the plurality of authorized entities, wherein the historical role-based pattern is indicative of a correlation between the ideal operations and the corresponding RBAC details; and training the anomaly detection model based on the historical role-based pattern to obtain the initial version of the anomaly detection model.

13. The method of claim 11, wherein analyzing the historical operation data comprises: identifying a historical behaviour-based pattern for the asset, wherein the historical behaviour-based pattern is indicative of a correlation between the one or more historical operations and the particular historical time; and optimizing the initial version of the anomaly detection model based on the historical behaviour-based pattern to obtain the final trained version of the anomaly detection model.

14. The method of claim 9, wherein the one or more preventive actions include at least one of: generating a suspension signal for transmission to one or more devices associated with the organization, wherein the suspension signal is to prevent execution of the one or more operations for which the anomaly is detected; and generating an alert notification for transmission to a supervisor on a supervisor device, wherein the alert notification is indicative of the anomaly.

15. The method of claim 9, wherein processing the real-time operation data and the RBAC information comprises: obtaining a historical role-based pattern for the asset, wherein the historical

role-based pattern is indicative of ideal operations performed by an ideal entity having authorization to operate within an ideal OT environment according to the authorization details and responsibilities; and comparing the ideal operations with the one or more operations to detect the anomaly, wherein detecting the anomaly comprises detecting a deviation of at least one operation of the one or more operations from the ideal operations.

16. A non-transitory computer-readable medium comprising instructions for detecting an anomaly in an operational technology (OT) environment, the instructions being executable by a processing resource to: obtain real-time operation data corresponding to a user functioning within an operational technology (OT) environment of an organization, the real-time operation data being indicative of one or more operations performed by the user within the OT environment of the organization; for each of the one or more operations, identify timing information associated with the operation, the timing information indicating a particular time at which the operation was performed; process, utilizing an anomaly detection model, the real-time operation data and the timing information to detect any deviation of the one or more operations from a historical operating pattern of the user in terms of the particular time; and initiate one or more preventive actions within the OT environment upon detecting a deviation of at least one of the one or more operations from the historical operating pattern.

17. The non-transitory computer-readable medium of claim 16, wherein the instructions are executable by the processing resource to: obtain, for the organization, historical operation data corresponding to one or more entities associated with the organization, wherein the historical operation data is indicative of one or more historical operations performed by each of the one or more entities within the OT environment of the organization; for each of the one or more historical operations, identify a particular historical time at which the historical operation was performed; and analyze the historical operation data in correlation with the particular historical time to obtain the anomaly detection model.

18. The non-transitory computer-readable medium of claim 17, wherein to analyze the historical operation data, the instructions are executable by the processing resource to: identify a historical behaviour-based pattern for the user, wherein the historical behaviour-based pattern is indicative of a correlation between the one or more historical operations and the particular historical time; and analyze the historical behaviour-based pattern to obtain the anomaly detection model.

19. The non-transitory computer-readable medium of claim 16, wherein the one or more preventive actions include at least one of: generating a suspension signal for transmission to one or more devices associated with the organization, wherein the suspension signal is to prevent execution of the at least one operation; and generating an alert notification for transmission to a supervisor on a supervisor device, wherein the alert notification is indicative of the at least one operation.

20. The non-transitory computer-readable medium of claim 16, wherein to process the real-time operation data and the timing information, the instructions are executable by the processing resource to: obtain a historical behaviour-based pattern of the user, wherein the historical behaviour-based pattern is indicative of historical operations performed by the user at the particular time; and compare the historical operations with the one or more operations to detect the deviation.
